Biotechnology and genetic modification

* Bacteria are useful in biotechnology and genetic modification due to their rapid reproduction rate and their ability to make complex molecules

Why are bacteria useful in biotechnology and genetic modification?
- Few ethical concerns over their manipulation and growth
- The presence of plasmids
* Small, no need for large space
* Has genetic material similar to other living organisms.

Bacteria is used to make:
1- Biofuels: Produced by fermenting large of sugar or grain, and then extracting and refining the ethanol.
- Ethanol releases fewer harmful emmissions than petrol
- Ethanol is renewable
* Biofuels are carbon neutral; CO_2 taken in by plants for photosynthesis is released back to atmosphere after combustion of biofuels
2- Bread making on yeast: The CO_2 produced by anaerobic respiration forms air bubbles in the dough, causing the bread to rise.

Anaerobic respiration: The chemical reaction in cells that break down nutrient molecules to release energy without using oxygen.

* Releases much less energy per glucose molecule than aerobic respiration.

- In yeast: Glucose ⟶ Alcohol + Carbon dioxide

$$C_6H_{12}O_6 \longrightarrow 2C_2H_5OH + 2CO_2$$

The effect of temperature in yeast production:

- Yeast respires anearobically.
- Yeast requires sugar to be activated in order to make anaerobic respiration
- Carbon dioxide produced will make the dough rise
- The process is controlled by enzymes. Enzymes work best from 35°-40°C, so you can increase the temperature up to 40°C to increase the respiration of the yeast
- If you increase the temperature higher than 40°C, the enzyme becomes denatured and respiration decreases

Pectinase in Fruit Juice Production

* Pectin is a substance that helps plant cells to stick together.

→ In order for a clear and a large yield of juice to be obtained from a fruit, pectin has to be broken down.

* Pectinease is used to break pectin between plact cells. Which produces a sweeter, clearer, and larger yield of juice

Enzymes in biological washing powders.

→ Protease enzymes are added to break down large insoluble proteins into smaller, soluble amino acids that can be washed away with water.

→ Lipase enzyme are added to break down large, insoluble fats into smaller, soluble fatty acids and glycerol that can be washed away with water

Lactase in producing lactose-free milk.

* Some people do not produce the enzyme lactase in their bodies, so they cannot digest lactose in diary products. People suffering from lactose-intolerance experience diarrhoea, bloating and cramps.
* Such people drink lactose free milk which is produced by adding lactase to the milk. Lactase breaks down lactose into glucose and galactose, making the milk suitable for people without them experiencing any symptoms

Penicillin

* Penicillin is an antibiotic used to treat a wide range of bacterial infections.

→ Penicillin is made by a fungus called pencillium, which is grown in fermenters.

* The fungus is grown in a culture medium containing carbohydrates (a source of glucose), which is used in respiration, and amine acids / ammonium salts (a source of nitrogen) used in making proteins and nucleic acids.

* Oxygen is supplied into the fermenter for aerobic respiration to take place in order for the funcus to release the energy used in producing penicillin.

→ pH has to be kept at a suitable level, usually 6-7, by adding acids and bases to the fermenter.

→ The fungus lives at room temperature, 24°-26°

→ penicillin is extracted by filtration

Single cell protein

- SCP is a type of food made by microorganisms

- SCP is sold as animal food because:
1- It tastes bitter
2- People are afraid of eating them

* One type of SCP is eaten by humans and it is called microprotein.
Made by a fungus

Advantages:
1- High fiber content
2- Can be made anywhere in fermenters
3- Low in fat and cholestrol content
4- Cheap
5- Better for animal welfare
6- Suitable for vegetarians and vegans

The Fermenter

PAWS

P: Probes: To monitor the temperature & pH

A: Air supply: Provides oxygen for aerobic respiration

W: Water jacket: Removes heat to maintain temperature at 24°-26°

S: Stirrer: Keeps the fungus in contact with nutrients & oxygen, and keeps an even temperature throughout the fermenter.

Conditions that should be controlled in a fermenter:

1- Temperature
2- pH
3- Oxygen
4- Nutrient supply
5- Waste products

Genetic modification: Changing the genetic material of an organism by removing, changing or inserting, individual genes.

Outline the process of genetic modification using bacterial production of a human protein:
1- Isolation of the DNA making up a human gene using restriction enzyme, forming sticky ends.
2- Cutting of bacterial plasmid DNA with the same restriction enzyme, forming complementary sticky ends.
3- Insertion of human DNA into bacterial plasmid using DNA ligase to form a recombinant plasmid
4- Insertion of recombinant plasmid into bacterial
5- multiplication of bacteria containing recombinant plasmids
6- Expression in bacteria of the human gene to make the human protein

Examples of genetic modification:
- The insertion of human genes into bacteria to produce human proteins
- The insertion of genes into crop plants to confer resistance to herbicides
- The insertion of genes into crop plants to confer resistance to insect pets
- The insertion of genes into crop plants to improve nutritional qualities

Advantages of genetic modification on crops (soya, maize, rice)
1- Reduced use of herbicides and pesticides: - Better for the environment
- Cheaper and less time consuming for farmers.
2- Increased crop yield: - Crops are not competing with weeds for resources
- Crops are not suffering from pest damage.
3- Enhanced nutritional content: - Crops have increased levels of vitamins and minerals
- May help to address nutrient deficiencies.
4- Draught and salinity tolerance: - Crops can be modified to tolerate draught and high soil salinity
- Allows for cultivation in regions with poor soil or water scarcity.

Disadvantages:
1- Gene transfer: Risks of resistant genes being transferred to wild plants by wind pollinators
2- Health concerns: Genetically modified crops may produce toxins or cause allergic reactions
3- Loss of biodiversity: Some variaties dominate increased vulnerability of agriculture to disease or environmental changes
4. Cost and control: - Companies that develo enetic modified seeds rally manipulate prices.
- Small farmers may be unable to combete with big, agricultural firms.